Deep Learning-based Sentiment Analysis in Persian Language

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2021 7th International Conference on Web Research (ICWR) | 978-1-6654-0426-6/20/$31.00 ©2021 IEEE | DOI: 10.1109/ICWR51868.2021.9443152

***Abstract*— Recently, interests in the appliance of deep learning techniques in natural language processing tasks considerably increased. Sentiment analysis is one of the most difficult tasks in natural language processing, mostly in the Persian Language. Thousands of websites, blogs, social networks like Telegram, Instagram and Twitter update, and modify by Persian users around the world that contains millions of contexts. To extract knowledge of these huge amounts of raw data, Deep Learning techniques became increasingly popular but there is a number of challenges that the novel models encounter with them. In this research, a hybrid deep learning-based sentiment analysis model proposed and implemented on customer reviews data of Digikala Online Retailer website. We already applied the classifier based on various deep learning networks and regularization techniques. Finally, by utilizing a hybrid approach, we achieved the best performance of 78.3 of F_1 score on three different classes: positive, negative, and neutral.**

Keywords*— *Deep learning, Natural Language Processing, Sentiment Analysis, Persian Language.

I. Introduction

In this day and age, with the increase of users access to various web-based communication platforms such as social networks, reviews and sales sites, the textual data produced in these platforms is growing exponentially. Analyzing this textual data is practical and useful in many ways. Today, various applications have been defined for the sentiment analysis in different fields such as politics and business intelligence. One of the largest online sales sites in Iran is the Digikala site, where users, after purchasing and evaluating the product, rate the product from different aspects and share comments for the use of others. Currently, the site is one of the richest textual resources for the field of natural language processing and text mining in Persian language.

II. Background

Since the deep learning-based sentiment analysis is a novel approach, there are few studies about the topic. Roshanfekr et al.[1] presented electronic products reviews dataset in Persian language and implemented deep learning techniques on it. Dashtipour et al.[2] developed a composition of Deep Auto Encoders, Deep CNN and utilized them to a novel Persian Films comments dataset. Their results show increased efficiency over Multi-Layer Perceptron with the accuracy of 82.86%. BokaeeNezhad et al.[3] suggested a new combined deep learning architecture for sentiment analysis in Persian language. Local features extracted by CNN[1] and long-term dependencies are learned by LSTM[2]. They also used Word2vec towards word representation as an unsupervised phase. They reached accuracy of 85%. Karimi et al.[4] evaluated attention models and BERT[3] as a famous pre-trained language representation in Persian. The reason of utilizing BERT is lack of enough data to achieve acceptable word representation vectors. They improved result of Learning Word Vectors for Sentiment Analysis paper[5] and demonstrated BERT works well than skip-gram, LSTM and CNN. Zobeidi etl al.[6] proposed a sentence level opinion classifier based on deep learning techniques. Their feature extraction part initiated by the help of Word2Vec and CNN. Bi-LSTM[4] network used for sentiment classification. They achieved 95% accuracy. Dashtipour et al. suggested a novel hybrid system for Persian Sentiment Analysis at concept level by Integrating Dependency Grammar Based Rules and DNN. They used Persian hotel reviews dataset as the benchmark. They achieved 86.29% accuracy. Rohanian et al.[7] proposed a new Persian sentiment analysis method for sentences classification into two and five by applying a convolution layer over input text data via various filters. Three distinct datasets were used to validate their model by AUC metric. Their results infer advantage of using CNN over compared to machine learning algorithms for Persian texts sentiment categorization mainly for small text data. Dastgheib et al.[8] suggested a hybrid method by composition of SCL[5] and CNN. The SCL method chooses most useful pivot attributes so the adaptation from one domain to same ones cannot drop the efficiency. Their result infers that initiating SCL+CNN hybrid approach can improve sentiment classification task for two domains over 10%. We already listed the recent research and studies of deep learning-based sentiment analysis on Persian texts.

III. Dataset

The current dataset is divided into 14 main categories as presented in Table III. In addition, the data are divided into different class categories. For example, in the data with classes, we have positive and negative class. We have three classes, one positive, one neutral and one negative, and so on. The dataset used in this research is supported by the Digikala Online Retailer Open Data Mining Program[6] including 100000 customer reviews on the different products categories as stated in Table I. All comments are in Persian Language. The Dataset contains following fields: *Product ID, Product Title, Title in English, User ID, Likes, Dislikes, Verification Status, Recommend, Title, Comment,*

[1] Convolutional Neural Network

[2] Long short-term memory
[3] Bidirectional Encoder Representations from Transformers
[4] Bidirectional Long Short-Term Memory
[5] structural correspondence learning
[6] Digikala.com/OpenData/

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Advantages and Disadvantages. Data group by text length bar chart is shown in Fig. 1.

A. Normalization

We read the user's comments file and remove the extra columns and convert the labels in binary shape to 0 and 1. Then we reshape the words in terms of writing. This includes the following steps:

1) remove punctuations,
2) remove characters and numbers,
3) remove stop words.

Regex corrects all these three steps[7].

B. Tokenization

After normalizing the text, the sentences must transform into the vectors, and each word must be converted into a token. NLTK library initiates tokenization phase from scratch.

C. Sentence Length Unification

At this point, the length of the sentences examined in terms of the number of tokens. Some sentences are even less than five words long. Some sentences exceed twenty words. According to our lookup, 90% of sentences contains fifteen words or less in length. Therefore, we decide Set the threshold to fifteen and select only the first fifteen words for sentences that are longer.

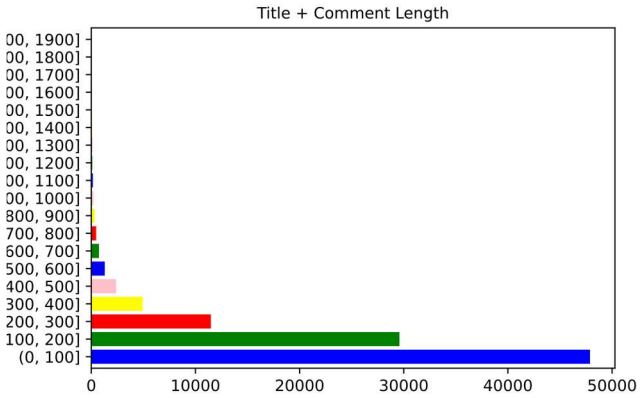

Fig. 1. Group by Data by Text Length

TABLE I. SAMPLE USER'S COMMENTS ABOUT DIFFERENT PRODUCTS CATEGORIES

Category	*User's Comments*
IT	واقعا عالیه، من که ازش خیلی راضیم
Home Appliance	گیره های فلزی پولیشر خیلی سخت تا می شوند و لذا حوله خیلی سخت در میاد
Mobile	گوشی با این قیمت و امکانات در مقایسه با سایر برندها هیچ مزیتی نداره.
Trimming Machine	این ماشین قیمت فوق العاده رقابتی داره، آدمو ترغیب میکنه برای خرید این محصول نکته ی دیگه، من با این قیمت انتظار نداشتم اصلاح به این منظمی رو داشته باشه اگر سیستم کارکردش هم از طریق باطری و هم مستقیم از طریق برق بود با این رده قیمتی عالی میشد
Player	بعضیها میگفتن این پخش ضعیف تر از نسخه های سیدی خور هست اما با تحقیقاتی ک داشتم گفتن تفاوتی نداره و تفاوتش در سیدی خور بودن و تعداد خروجی هاست. اگه یه سیستم میانرده میخواین و تعداد خروجی زیاد براتون‌مهم نیست، تو این رنج قیمت بهترین‌گزینه هست.

D. Vectorization

There are different ways to convert words to vectors, Including Term-Frequency, TF-IDF and more[9][10]. One of the famous approaches towards semantic relationship between words is the method introduced by Mikolov in 2013[11][12]. Obviously, the more data we have, the better the semantic relationship between words will be available. That is why we use pre-trained datasets. One of the famous pre-trained dataset is GloVe[13]. It is trained on 6 billion tokens and contain over 400,000 different words that Released in various 50, 100, 200, and 300 dimensions. Using the pre-trained GloVe model, we convert words into vectors of length 50. It should be noted that some words due to incorrect typing in the model are missing value and are filled with 0.

E. Train / Test Data Spliting

The dataset contains user's comments about different products of Digikala online Retailer website. After shuffling all the sentences, we split 80% of the dataset for training and 20% of the data for test data. Due to the heavy dimensions and volume of data, the RAM[8] was not enough sufficient; we split the data into 200,000 batches and store it on disk. In training state, we load the data batch in memory, respectively.

IV. NETWORK ARCHITECTURE AND TRAINING

In this section, we examine different architectures and techniques in neural network training to achieve the best accuracy. This is the simplest type of network. It has no hidden layer and has SoftMax as the output layer.

A. Network with non-hidden layer and SoftMax as output layer

The inputs are the vectors of the sentences. These inputs can also assign as batch to the network. The size of the input dimension is as (1):

$$Batch\ Size * 15 * 50 \quad (1)$$

Indicates 15 words per sentence and 50 dimensions per word. The output dimension is 2. It means the output is either 01 or 10.

B. Loss Function

The Loss calculation function is also an entropy function that is calculate as (2):

$$cross_entropy = -tf.reduce_mean(Y_ * tf.log(Y)) \quad (2)$$

$Y_$ is the main output and $Y_$ is the activity function. The mean multiplication of this value is equal to the error as (3):

$$H(p,q) = -\sum_{x \in X} p(x) \log q\,(x) \quad (3)$$

[7] Regular Expressions

[8] Random Access Memory

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According to studies, this error function works better than other functions such as **MSE**[9].

C. Activation Function

The activity function is calculated as follows. When the input vector multiplied by weights and multiplied by bias as (4)

$$Y = tf.nn.softmax(tf.matmul(XX,W) + b) \quad (4)$$

In addition, this value passed to the SofMax function as (5)

$$Softmax(L_n) = \frac{e^{L}n}{||e^{L}||} \quad (5)$$

D. Optimization algorithm and song learning

The algorithm used here is gradient reduction and the learning rate is 0.003.

V. RESULTS

In This section, we represent the models utilized in this research along with their final information values. Due to the large volume and the complexity of the data, strong hardware system needed for this study. We utilized a system with the following configuration:

1) CPU Intel corei7 8750 4 GHZ 2) RAM: 16G DDR4 3) Hard Disk: SSD M2 and 4) GPU: GeForce GTX 1060. Network's Data Values is shown in Table II, User's comments statistics is shown in Table III and Performance of Models on Digikala Dataset is shown in Table IV.

A. Word2Vec plus SoftMax

The network output after 100 epochs is shown. The best accuracy achieved in this network is 63.1, which was reached in the 13th Epoch. It shows that learning in this network is slow and it doesn't learn well.

B. MLP Network with 5 Hidden Layers and Sigmoid Activation Function

Five hidden layers added to the previous network. The activation function of the layers is sigmoid. The error optimization function is Adam. Number of hidden neurons are 200, 100, 60 and 30. Activation function of output layer is Softmax and the other parameters are as the same as the first network. The condition for learning decay is lack of improving the accuracy after 27 epochs. After 27 epochs, network accuracy increased which is improvement compared to the pervious network, but overfitting is still a challenging problem.

C. MLP Network with 5 Hidden Layers and ReLU Activation Function and LR Decay

The parameters are the same as the pervious network except the Reul activation function. Given the type of this network function, it has more power to jump out of local optimizations. Also, the learning rate is not constant, and decreases based on the current epoch number. Here's the formula (6) for calculating the learning rate:

$$0.0001 + 0.003 * (1/e)^{(step/2000)} \quad (6)$$

[9] Mean squared error.

TABLE II. NETWORK'S DATA VALUES

Data	*Value*
Batch Size	512
Learning Rate	0.001
Learning Rate Decay	0.9
Drop out	0.5
Number of LSTM cell for each word	100
Number of LSTM cell for each Character	50

TABLE III. USER'S COMMENTS STATISTICS

Category	*Positive*	*Negative*	*Neutral*
IT (General)	2673	803	537
Digital Products	6435	3885	1768
Home Appliances	4183	1922	1446
Mobile	546	107	92
Beauty, Health	4142	1803	1360
Personal Assistant	2243	824	504
Cars, Tools, Office Supplies	3687	1513	1030
Kitchen Appliances	3070	1369	927
Books, Stationery and Art	2171	746	561
Electric (Home)	1543	439	365
Audiovisual Equipment	1167	418	252
Fashion and Apparel	4057	1831	1408
Health, Detergents, Perfumes	399	142	96
Baby & Child Health	166	69	39

TABLE IV. PERFORMANCE OF MODELS ON DIGIKALA DATASET

Model	*F1*
Word2Vec+SoftMax	63.1
Word2Vec+5Layers+Sigmoid	72
Word2Vec+5Layers+ReLU+lrDecay	72.1
Word2Vec+5Layers+ReLU+lrDecay+DropOut	71.8
Word2Vec+LSTM	75.7
CharEmbed+Word2Vec+LSTM+RUS	73.9
CharEmbed+Word2Vec+LSTM	**78.3**

The step refers to the epoch number and the range is from 0.0031 to 0.0001. after 5 epochs the network reached 72.01 of accuracy which is a slight improvement over the previous one. But the overfitting challenge remains a major problem.

D. MLP Network with 5 Hidden Layers, ReLU Activation Function, LR Decay and Dropout technique

The parameters are the same as the pervious network. Only the dropout technique has been added to the current network by removing 0.25 neurons and all edges attached to each hidden layer. after 5 epochs the network reached 72.01 of accuracy That has dropped slightly over the previous one. But the highlight is handling the network overfitting thanks to the dropout technique.

E. Long Short-Term Memory Network

We remove the hidden layers of the previous network and place an LSTM layer with 100 hidden neurons instead. After 9 epochs, the network reached 75.07 which is a remarkable increase in accuracy thanks to the intrinsic structure of recursive networks that take into account the role of words in a sentence.

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F. Bidirectional recurrent neural networks – BiLSTM

After 14 epochs the network reached 78.03 which is almost equal to the precision of the previous network.

G. Gated Recurrent Unit – GRU Network

The GRU network has a simpler structure than the two previous networks, so we can see that the network output is less accurate. After 6 epochs the network reached accuracy of 73.09 which is less accurate than the LSTM network family.

H. Most important causes of people's satisfaction and dissatisfaction about products:

- **Ngrams**

The most fundamental language model is the n-gram. In the fields of computational linguistics and probability, n-gram is a continuous sequence of N words. The main idea behind n-gram language model is to collect statistical information about the frequency of different n-grams. The result is capable to enhance the forecasting of QAC[10]. N-grams visualization is presented in Fig 2, 3, 4 and Fig.5.

Fig. 2. users satisfaction reason with digital goods category

Fig. 3. users dissatisfaction reasons with digital goods category

Fig. 4. users satisfaction reasons with Home and kitchen goods category

Fig. 5. users dissatisfaction reason with the Home and kitchen goods category

[10] Query Auto-Completion

VI. Conclusion and Future Works

In this research, we proposed a hybrid deep learning-based sentiment analysis model and applied it on a persian customer reviews dataset of Digikala Online Retailer website. One of the highlights of this study is implementation of the proposed model on three classes of positive, negative and neutral, unlike many related studies that were conducted sentiment analysis approach on just two positive and negative classes. Although we faced two limitations which are our main challenges in conducting and developing research to increase accuracy: 1) The need for a large Persian training dataset and 2) access to powerful GPUs. The motivation of our research is employment of deep learning-based approach by the different related models and techniques on Persian Language. Certainly, Sentiment analysis approach needed for the Persian language, but it has its own challenges, both conversational and formal. It is also possible to achieve higher accuracy due to hybrid techniques found in new articles such as combining char embedding with word embedding. Regarding to the growing interests in application of advanced deep learning techniques in natural language processing tasks, we will utilize state-of-the-arts transformer-based techniques on different Persian datasets.

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